

PHYSICS CURRICULUM SYLLABUS

Complete O'Level & A'Level Modular Course Outline

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Curriculum Structure & Overview: This document presents a rigorous, structured physics curriculum divided into two clear academic tiers: **Part 1 (O'Level / GCSE / IGCSE equivalent)** focusing on foundational physical laws, qualitative reasoning, and core algebra, and **Part 2 (A'Level / AS & A2 equivalent)** expanding into calculus-based dynamics, advanced field theories, wave superposition, AC circuits, thermal physics, and quantum phenomena.

PART 1: O'LEVEL PHYSICS CURRICULUM (FOUNDATIONAL)

1. General Physics & Mechanics

• 1.1 Physical Quantities, Units & Measurement

- SI base units (m, kg, s, A, K, mol, cd) and derived units
- Vector and scalar quantities
- Measurement instruments: micrometers, vernier calipers, stopwatches
- Precision, accuracy, systematic errors, and random errors

• 1.2 Motion (Kinematics)

- Distance, displacement, speed, velocity, and acceleration
- Distance-time and velocity-time graphs (calculating gradients and area under curves)
- Equations of uniform acceleration: $v = u + at$, $s = ut + \frac{1}{2}at^2$, $v^2 = u^2 + 2as$
- Free-fall acceleration due to gravity and terminal velocity

• 1.3 Forces & Dynamics

- Newton's Laws of Motion (First, Second, and Third Laws)
- Resultant forces, mass, weight, and gravitational field strength ($W = mg$)
- Friction, drag forces, and air resistance
- Qualitative treatment of circular motion and centripetal force

• 1.4 Turning Effects & Momentum

- Moment of a force ($Moment = F \times d$) and Principle of Moments
- Center of gravity and stability of structures
- Linear momentum ($p = mv$) and impulse ($F\Delta t = \Delta p$)
- Conservation of linear momentum in 1D collisions and explosions

• 1.5 Work, Energy, & Power

- Work done ($W = Fd$) and energy transfer
- Kinetic energy ($E_k = \frac{1}{2}mv^2$) and Gravitational potential energy ($E_p = mgh$)
- Law of Conservation of Energy and energy dissipation
- Power ($P = W/t$) and percentage efficiency

• 1.6 Hydrostatics & Fluid Statics (Equilibrium of Fluids)

- Pressure on solids ($P = F/A$)
- Hydrostatic pressure in fluids ($P = \rho gh$)
- **Equilibrium of fluids:** Pascal's Principle and hydraulic press applications
- Archimedes' Principle, buoyant force, and conditions for flotation
- Atmospheric pressure, barometers, and manometers

2. Thermal Physics

• 2.1 Kinetic Model of Matter & Gas Laws

- States of matter (solids, liquids, gases) and molecular arrangements
- Brownian motion and diffusion as evidence for kinetic theory
- Gas pressure and temperature explained by molecular collision dynamics
- Boyle's Law ($P_1V_1 = P_2V_2$), Charles's Law, and Pressure Law

• 2.2 Thermal Properties & Temperature

- Thermal expansion of solids, liquids, and gases
- Thermometric properties and temperature scales (Celsius and Kelvin)
- Heat capacity and Specific Heat Capacity ($Q = mc\Delta T$)
- Latent heat and Specific Latent Heat of fusion and vaporization ($Q = mL$)

• 2.3 Thermal Energy Transfer

- Conduction: free electron diffusion and lattice vibration
- Convection: fluid density changes and circulation currents
- Radiation: infrared radiation, emission, absorption, and surface characteristics
- Qualitative introduction to thermal conduction rate

3. Waves & Optics

• 3.1 Wave Properties & Behavior

- Transverse vs. longitudinal waves
- Wave parameters: amplitude, wavelength (λ), frequency (f), period (T), and velocity ($v = f\lambda$)
- Fundamental wave behaviors: reflection, refraction, and diffraction

• 3.2 Light & Optics

- Reflection in plane mirrors and laws of reflection
- Refraction, Snell's Law ($n = \sin i / \sin r$), critical angle, and Total Internal Reflection (TIR)
- Thin converging and diverging lenses (ray diagrams, focal length, real vs. virtual images)
- Electromagnetic (EM) spectrum: order, properties, uses, and safety hazards

• 3.3 Sound

- Production and longitudinal propagation of sound waves
- Speed of sound across solid, liquid, and gas media
- Pitch (frequency) and loudness (amplitude)
- Echoes, sonar, and ultrasound applications

4. Electricity & Magnetism

• 4.1 Static Electricity

- Electric charges, charging by friction, conduction, and induction
- Electric field lines around point and surface charges

• 4.2 Current Electricity

- Current ($I = Q/t$), Potential Difference ($V = W/Q$), and Resistance ($R = V/I$)
- Ohm's Law and I-V characteristics (ohmic conductors, filament lamps, diodes)
- Resistivity factors (length, cross-sectional area, material)
- Electrical energy ($E = VIt$) and power ($P = VI = I^2R = V^2/R$)

• 4.3 DC Circuits

- Series and parallel circuit rules for current, voltage, and total resistance

- Potential divider circuits, LDRs, and thermistor sensors
- Electrical safety: fuses, circuit breakers, earthing, double insulation

• 4.4 Magnetism & Electromagnetism

- Properties of permanent magnets and magnetic field lines
- Magnetic fields around straight wires, flat coils, and solenoids
- Force on current-carrying conductors (Fleming's Left-Hand Rule, DC motor)
- Electromagnetic induction (Faraday's Law, Lenz's Law, AC generator, Transformers)

5. Nuclear & Space Physics

• 5.1 Atomic & Nuclear Structure

- Rutherford's α -scattering experiment and planetary model of the nucleus
- Protons, neutrons, electrons, isotopes, and standard nuclide notation (A_ZX)

• 5.2 Radioactivity

- Alpha (α), Beta (β), and Gamma (γ) decay, ionization energy, and penetration power
- Radioactive decay equations and background radiation compensation
- Half-life ($t_{1/2}$) determination and decay curves
- Practical applications and safety hazards of radioisotopes

• 5.3 Space Physics (Introductory)

- Solar system structure, orbital mechanics, and gravitational attraction
- Stellar life cycles: main sequence, red giants, white dwarfs, supernovas, neutron stars/black holes
- Cosmology: Light-years, Hubble redshift, Big Bang Theory evidence

PART 2: A'LEVEL PHYSICS CURRICULUM (ADVANCED)

1. Advanced Mechanics, Fluids & Materials (AS Level)

• 1.1 Physical Quantities & Measurement Analysis

- Homogeneity of physical equations using SI base units
- Vector addition, subtraction, and 2D/3D component resolution
- Absolute, fractional, and percentage uncertainties; error propagation

• 1.2 Kinematics & Dynamics

- Two-dimensional kinematics: Projectile motion analysis
- Calculus formulation of motion ($\mathbf{v} = d\mathbf{s}/dt$, $\mathbf{a} = d\mathbf{v}/dt$, $\mathbf{s} = \int \mathbf{v} dt$)
- 2D momentum conservation, elastic vs. inelastic collisions, center of mass frame

• 1.3 Statics & Equilibrium

- Torque, moments, and couple forces
- Rigid body equilibrium conditions ($\Sigma \mathbf{F} = \mathbf{0}$, $\Sigma \boldsymbol{\tau} = \mathbf{0}$)

• 1.4 Elasticity & Mechanical Moduli

- Stress (σ), strain (ϵ), and Young's Modulus ($E = \sigma/\epsilon$)
- Hooke's Law, elastic limit, yield point, plastic deformation, and strain energy density
- **Bulk Modulus (K)**: Volumetric stress/strain and fluid compressibility ($K = -V \Delta P / \Delta V$)
- **Shear Modulus (G)**: Shear stress/strain and rigidity ($G = \tau / \gamma$)
- **Poisson's Ratio (ν)**: Ratio of lateral strain to axial strain ($\nu = -\epsilon_{trans} / \epsilon_{axial}$)

• 1.5 Fluid Dynamics & Surface Phenomena

- **Viscosity**: Laminar flow, coefficient of viscosity (η), and Newton's law of viscosity ($\mathbf{F} = \eta A dv/dy$)
- **Stokes' Law**: Viscous drag force on spherical bodies ($\mathbf{F} = 6\pi\eta r \mathbf{v}$) and terminal velocity
- **Surface Tension (γ)**: Cohesive/adhesive forces, surface free energy
- Capillary action, contact angle, and excess pressure across curved interfaces ($\Delta P = 2\gamma/r$)

2. Waves, Electricity, & Particle Physics (AS Level)

• 2.1 Advanced Waves & Wave Superposition

- **Progressive Waves**: Wave function $y = A \sin(\omega t - kx)$, phase difference, and path difference
- **Wavefronts**: Huygens' principle, plane and spherical wavefront propagation
- **Stationary (Standing) Waves**: Principle of superposition, nodes, antinodes, string harmonics, and resonant air columns
- Interference: Young's Double Slit experiment ($\lambda = ax/D$), multi-slit diffraction gratings ($d \sin \theta = n\lambda$)
- Doppler Effect for moving sound sources and astronomical light sources

• 2.2 Advanced Electricity

- Internal resistance, terminal potential difference, and electromotive force ($E = I(R + r)$)
- Kirchhoff's First Law (conservation of charge) and Second Law (conservation of energy)
- Potentiometer circuits, Wheatstone bridge, and sensor calibration

• 2.3 Particle & Quantum Physics Foundations

- Classification of subatomic particles: Hadrons (Baryons, Mesons), Leptons, and Quarks (up, down, strange)
- Fundamental forces (Strong, Electromagnetic, Weak, Gravitational) and exchange bosons
- Photon energy ($E = hf$), photoelectric effect, work function (Φ), stopping potential, and Einstein's equation
- Wave-particle duality and de Broglie wavelength ($\lambda = h/p$)

3. Advanced Mechanics, Fields & Thermodynamics (A2 Level)

• 3.1 Circular Motion & Gravitational Fields

- Angular velocity (ω), centripetal acceleration ($a = \omega^2 r = v^2/r$), and centripetal force
- Newton's Law of Universal Gravitation ($F = G m_1 m_2 / r^2$)
- Gravitational field strength (g) and Gravitational Potential ($V_g = -GM/r$)
- Kepler's Laws of planetary motion, geostationary orbits, and orbital escape velocity

• 3.2 Simple Harmonic Motion (SHM)

- SHM differential equation condition ($a = -\omega^2 x$)
- Kinematic equations for displacement, velocity, and acceleration
- Energy transformation between kinetic and potential energy in SHM
- Damping (light, heavy, critical), forced oscillations, and resonance curves

• 3.3 Thermal Physics, Heat Transfer, & Thermodynamics

- Ideal gas equation ($pV = nRT = N k_B T$) and kinetic theory derivation ($pV = \frac{1}{3} N m \langle c^2 \rangle$)
- **Fourier's Law of Thermal Conduction:** Heat transfer rate ($dQ/dt = -kA dT/dx$) and thermal conductivity (k)
- First Law of Thermodynamics ($\Delta U = Q - W$), isothermal, isobaric, isochoric, and adiabatic processes
- **Second Law of Thermodynamics:**
 - Clausius and Kelvin-Planck statements
 - Heat engines, thermal pumps, and refrigeration cycles
 - Carnot engine efficiency ($\eta = 1 - T_C/T_H$)
 - Entropy (S), microscopic disorder, and thermodynamic irreversibility

4. Electricity, Magnetic Fields & AC Circuits (A2 Level)

• 4.1 Electric Fields & Capacitance

- Coulomb's Law ($F = q_1 q_2 / (4\pi\epsilon_0 r^2)$)
- Electric field intensity (E) and Electric Potential (V) around point charges and parallel plates
- Capacitance ($C = Q/V$), parallel plate formula, and dielectric permittivity (ϵ)
- Capacitor energy storage and RC transient charging/discharging exponential decay ($\tau = RC$)

• 4.2 Magnetic Fields & Electromagnetic Induction

- Magnetic flux density (B), flux (Φ), and flux linkage ($N\Phi$)
- Lorentz force on charged particles ($F = Bqv \sin \theta$) and current conductors ($F = BIL \sin \theta$)
- **Cyclotron Mechanics:** Motion of charged particles in magnetic fields, circular path radius ($r = mv / qB$), cyclotron frequency ($f = qB / 2\pi m$), and particle acceleration in D-shaped electrodes
- Hall Effect mechanics and Hall voltage derivation
- Faraday's Law and Lenz's Law ($\epsilon = -N d\Phi/dt$)
- Self-inductance (L) and mutual inductance

• 4.3 Alternating Current (AC) Circuits

- Sinusoidal AC sources, peak values, and root-mean-square metrics ($V_{rms} = V_0 / \sqrt{2}$)
- **Reactance:** Inductive reactance ($X_L = \omega L$) and Capacitive reactance ($X_C = 1 / (\omega C)$)
- **Impedance:** Series RLC circuit analysis, phase angles, and phasor diagrams ($Z = \sqrt{R^2 + (X_L - X_C)^2}$)
- **Resonance:** Resonant condition ($X_L = X_C$), resonant frequency ($f_0 = 1 / (2\pi\sqrt{LC})$), Quality factor (Q), and average power factor ($\cos \phi$)
- Power dissipation, transformers, and AC rectification (half-wave and full-wave bridge)

5. Modern, Nuclear, & Astrophysics (A2 Level)

• 5.1 Quantum Mechanics & Atomic Physics

- Atomic energy levels, line emission, and line absorption spectra
- X-ray production mechanism, characteristic radiation, and attenuation laws in medical imaging

• 5.2 Nuclear Physics

- Mass defect and Nuclear Binding Energy ($\Delta E = \Delta m c^2$)
- Binding energy per nucleon curve and nuclear stability criteria
- Nuclear fission and thermonuclear fusion processes
- Radioactive decay law ($N = N_0 e^{-\lambda t}$) and decay constant relation ($\lambda = \ln 2 / t_{1/2}$)

• 5.3 Astrophysics & Cosmology

- Stellar radii, total luminosity, Stefan-Boltzmann Law ($L = 4\pi r^2 \sigma T^4$), and Wien's Displacement Law
- Hertzsprung-Russell (H-R) diagram classification of stellar evolution
- Standard candles and Cepheid variable distance measurements
- Hubble's Law ($v = H_0 d$), expansion of space, Cosmic Microwave Background Radiation (CMBR), and Dark Energy